

NOV. 29,2003

[DATA FOR REFERENCE]



103AP-2

TEMPERATURE VS RESISTANCE CHARACTERISTICS [ITS-90]

Resistance 10.0k ohms at 25 Degree C
Resistance Tolerance $\pm 0.5\%$
B Value 3 435K at 25/85 Degree C
B Value Tolerance $\pm 0.5\%$

Ishizuka Electronics Corp
:7-7,KINSHI 1-CHOME,
SUMIDA-KU,TOKYO 130 JAPAN
TEL (3) 3621-2704
FAX (3) 3623-7776

Temp. (deg C)	Rmax. (k ohms)	Rst. (k ohms)	Rmin. (k ohms)	Tolerance (deg C)
-60	616.1	600.6	585.4	-0.40 + 0.41
-59	578.1	563.7	549.6	-0.40 + 0.40
-58	542.7	529.4	516.3	-0.40 + 0.40
-57	509.8	497.4	485.3	-0.40 + 0.40
-56	479.2	467.7	456.5	-0.40 + 0.40
-55	450.7	440.0	429.5	-0.40 + 0.40
-54	424.1	414.2	404.4	-0.40 + 0.40
-53	399.3	390.0	381.0	-0.39 + 0.40
-52	376.1	367.5	359.1	-0.39 + 0.40
-51	354.5	346.5	338.7	-0.39 + 0.39
-50	334.3	326.9	319.6	-0.39 + 0.39
-49	315.5	308.5	301.7	-0.39 + 0.39
-48	297.8	291.3	285.0	-0.39 + 0.39
-47	281.3	275.2	269.3	-0.39 + 0.39
-46	265.8	260.1	254.6	-0.38 + 0.39
-45	251.3	246.0	240.8	-0.38 + 0.38
-44	237.7	232.8	227.9	-0.38 + 0.38
-43	224.9	220.3	215.8	-0.38 + 0.38
-42	213.0	208.7	204.5	-0.38 + 0.38
-41	201.7	197.7	193.8	-0.38 + 0.38
-40	191.2	187.4	183.7	-0.37 + 0.36
-39	181.0	177.5	174.0	-0.36 + 0.36
-38	171.5	168.2	164.9	-0.36 + 0.36
-37	162.5	159.4	156.4	-0.36 + 0.36
-36	154.0	151.1	148.3	-0.36 + 0.36
-35	146.1	143.4	140.7	-0.35 + 0.35
-34	138.6	136.1	133.6	-0.35 + 0.35
-33	131.6	129.2	126.9	-0.35 + 0.35
-32	125.0	122.8	120.6	-0.35 + 0.35
-31	118.7	116.7	114.6	-0.34 + 0.35
-30	112.9	110.9	109.0	-0.34 + 0.34
-29	107.2	105.4	103.6	-0.33 + 0.33
-28	101.9	100.1	98.50	-0.33 + 0.33
-27	96.78	95.22	93.67	-0.33 + 0.33
-26	92.03	90.57	89.12	-0.33 + 0.33
-25	87.56	86.18	84.83	-0.32 + 0.32
-24	83.33	82.04	80.77	-0.32 + 0.32
-23	79.34	78.13	76.94	-0.32 + 0.32
-22	75.57	74.44	73.32	-0.32 + 0.32
-21	72.00	70.94	69.89	-0.31 + 0.31
-20	68.63	67.64	66.65	-0.31 + 0.31
-19	65.38	64.44	63.52	-0.30 + 0.30
-18	62.30	61.42	60.56	-0.30 + 0.30
-17	59.39	58.57	57.76	-0.30 + 0.30
-16	56.63	55.87	55.10	-0.29 + 0.29
-15	54.03	53.31	52.59	-0.29 + 0.29
-14	51.56	50.88	50.21	-0.29 + 0.29
-13	49.22	48.59	47.96	-0.29 + 0.29
-12	47.01	46.41	45.82	-0.28 + 0.28

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Temp. (deg C)	Rmax. (k ohms)	Rst. (k ohms)	Rmin. (k ohms)	Tolerance (deg C)	
-11	44.91	44.35	43.80	-0.28	+ 0.28
-10	42.92	42.39	41.87	-0.28	+ 0.27
-9	40.99	40.50	40.01	-0.27	+ 0.27
-8	39.17	38.70	38.24	-0.26	+ 0.27
-7	37.43	37.00	36.57	-0.26	+ 0.26
-6	35.79	35.38	34.98	-0.26	+ 0.26
-5	34.23	33.85	33.47	-0.26	+ 0.26
-4	32.75	32.39	32.03	-0.25	+ 0.25
-3	31.34	31.00	30.67	-0.25	+ 0.25
-2	30.00	29.69	29.37	-0.25	+ 0.25
-1	28.73	28.44	28.14	-0.24	+ 0.24
0	27.52	27.25	26.97	-0.24	+ 0.24
1	26.36	26.10	25.84	-0.23	+ 0.23
2	25.25	25.00	24.76	-0.23	+ 0.23
3	24.19	23.96	23.73	-0.23	+ 0.23
4	23.19	22.97	22.76	-0.22	+ 0.22
5	22.23	22.03	21.83	-0.22	+ 0.22
6	21.32	21.13	20.94	-0.22	+ 0.22
7	20.46	20.28	20.10	-0.21	+ 0.21
8	19.63	19.46	19.30	-0.21	+ 0.21
9	18.85	18.69	18.53	-0.20	+ 0.21
10	18.10	17.95	17.80	-0.20	+ 0.20
11	17.37	17.23	17.09	-0.19	+ 0.20
12	16.68	16.55	16.42	-0.19	+ 0.19
13	16.02	15.90	15.77	-0.19	+ 0.19
14	15.39	15.27	15.16	-0.18	+ 0.18
15	14.79	14.68	14.57	-0.18	+ 0.18
16	14.21	14.11	14.01	-0.18	+ 0.18
17	13.67	13.57	13.48	-0.17	+ 0.17
18	13.14	13.05	12.97	-0.17	+ 0.17
19	12.64	12.56	12.48	-0.16	+ 0.17
20	12.17	12.09	12.01	-0.16	+ 0.16
21	11.70	11.63	11.56	-0.15	+ 0.16
22	11.26	11.20	11.13	-0.15	+ 0.15
23	10.84	10.78	10.72	-0.15	+ 0.15
24	10.44	10.38	10.32	-0.14	+ 0.14
25	10.05	10.00	9.950	-0.14	+ 0.14
26	9.683	9.633	9.582	-0.14	+ 0.14
27	9.332	9.281	9.231	-0.15	+ 0.15
28	8.995	8.945	8.895	-0.16	+ 0.16
29	8.673	8.623	8.573	-0.16	+ 0.16
30	8.364	8.314	8.264	-0.17	+ 0.17
31	8.065	8.016	7.966	-0.17	+ 0.17
32	7.779	7.730	7.681	-0.18	+ 0.18
33	7.504	7.456	7.407	-0.18	+ 0.19
34	7.241	7.193	7.145	-0.19	+ 0.19
35	6.989	6.941	6.894	-0.20	+ 0.20
36	6.747	6.700	6.653	-0.20	+ 0.20
37	6.515	6.468	6.422	-0.21	+ 0.21
38	6.293	6.246	6.200	-0.22	+ 0.22
39	6.079	6.033	5.987	-0.22	+ 0.22

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Temp. (deg C)	Rmax. (k ohms)	Rst. (k ohms)	Rmin. (k ohms)	Tolerance (deg C)	
40	5.874	5.829	5.783	-0.23	+ 0.23
41	5.675	5.630	5.585	-0.23	+ 0.23
42	5.484	5.440	5.395	-0.24	+ 0.24
43	5.300	5.257	5.213	-0.25	+ 0.25
44	5.124	5.081	5.038	-0.25	+ 0.25
45	4.955	4.912	4.870	-0.26	+ 0.26
46	4.792	4.750	4.708	-0.27	+ 0.27
47	4.636	4.594	4.553	-0.27	+ 0.27
48	4.485	4.444	4.404	-0.28	+ 0.28
49	4.340	4.300	4.260	-0.29	+ 0.29
50	4.201	4.162	4.122	-0.29	+ 0.29
51	4.066	4.027	3.988	-0.30	+ 0.30
52	3.936	3.897	3.859	-0.30	+ 0.30
53	3.811	3.773	3.735	-0.31	+ 0.31
54	3.690	3.653	3.616	-0.32	+ 0.32
55	3.574	3.537	3.501	-0.32	+ 0.32
56	3.462	3.426	3.391	-0.33	+ 0.33
57	3.355	3.319	3.284	-0.34	+ 0.34
58	3.251	3.216	3.182	-0.34	+ 0.35
59	3.152	3.117	3.083	-0.35	+ 0.35
60	3.055	3.022	2.988	-0.36	+ 0.36
61	2.962	2.929	2.896	-0.36	+ 0.36
62	2.872	2.839	2.807	-0.37	+ 0.37
63	2.785	2.753	2.721	-0.38	+ 0.38
64	2.701	2.670	2.638	-0.38	+ 0.39
65	2.621	2.589	2.559	-0.39	+ 0.39
66	2.543	2.512	2.482	-0.40	+ 0.40
67	2.468	2.438	2.408	-0.41	+ 0.41
68	2.395	2.366	2.336	-0.41	+ 0.41
69	2.325	2.296	2.267	-0.42	+ 0.42
70	2.258	2.229	2.201	-0.43	+ 0.43
71	2.192	2.164	2.136	-0.43	+ 0.43
72	2.128	2.101	2.074	-0.44	+ 0.44
73	2.067	2.040	2.013	-0.45	+ 0.45
74	2.008	1.981	1.955	-0.45	+ 0.46
75	1.951	1.925	1.899	-0.46	+ 0.46
76	1.896	1.870	1.845	-0.47	+ 0.47
77	1.842	1.817	1.792	-0.48	+ 0.48
78	1.791	1.766	1.741	-0.48	+ 0.49
79	1.741	1.716	1.692	-0.49	+ 0.49
80	1.693	1.669	1.645	-0.50	+ 0.50
81	1.646	1.622	1.599	-0.50	+ 0.51
82	1.600	1.577	1.555	-0.51	+ 0.51
83	1.557	1.534	1.512	-0.52	+ 0.52
84	1.514	1.492	1.470	-0.53	+ 0.53
85	1.473	1.451	1.430	-0.54	+ 0.54
86	1.433	1.412	1.391	-0.54	+ 0.54
87	1.395	1.374	1.353	-0.55	+ 0.55
88	1.358	1.337	1.316	-0.56	+ 0.56
89	1.321	1.301	1.281	-0.57	+ 0.57
90	1.286	1.266	1.247	-0.57	+ 0.57

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<u>Temp. (deg C)</u>	<u>Rmax. (k ohms)</u>	<u>Rst. (k ohms)</u>	<u>Rmin. (k ohms)</u>	<u>Tolerance (deg C)</u>	
91	1.252	1.233	1.213	-0.58	+ 0.58
92	1.220	1.200	1.181	-0.59	+ 0.59
93	1.188	1.169	1.150	-0.60	+ 0.60
94	1.157	1.138	1.120	-0.60	+ 0.61
95	1.127	1.108	1.090	-0.61	+ 0.62
96	1.098	1.080	1.062	-0.62	+ 0.62
97	1.070	1.052	1.035	-0.63	+ 0.63
98	1.042	1.025	1.008	-0.64	+ 0.64
99	1.016	0.9990	0.9825	-0.65	+ 0.65
100	0.9901	0.9737	0.9575	-0.65	+ 0.65
101	0.9651	0.9490	0.9331	-0.66	+ 0.66
102	0.9408	0.9250	0.9095	-0.67	+ 0.67
103	0.9173	0.9018	0.8865	-0.67	+ 0.68
104	0.8945	0.8793	0.8643	-0.68	+ 0.69
105	0.8724	0.8574	0.8427	-0.69	+ 0.70
106	0.8510	0.8363	0.8218	-0.70	+ 0.70
107	0.8302	0.8157	0.8015	-0.71	+ 0.71
108	0.8100	0.7958	0.7818	-0.72	+ 0.72
109	0.7904	0.7764	0.7627	-0.73	+ 0.73
110	0.7713	0.7576	0.7441	-0.74	+ 0.73
111	0.7527	0.7392	0.7260	-0.74	+ 0.74
112	0.7347	0.7214	0.7084	-0.75	+ 0.75
113	0.7171	0.7041	0.6913	-0.76	+ 0.76
114	0.7001	0.6873	0.6747	-0.77	+ 0.77
115	0.6836	0.6710	0.6586	-0.78	+ 0.78
116	0.6675	0.6552	0.6430	-0.78	+ 0.79
117	0.6519	0.6398	0.6278	-0.79	+ 0.80
118	0.6367	0.6248	0.6131	-0.80	+ 0.81
119	0.6220	0.6103	0.5987	-0.81	+ 0.82
120	0.6077	0.5961	0.5848	-0.82	+ 0.82
121	0.5937	0.5823	0.5712	-0.83	+ 0.83
122	0.5801	0.5689	0.5580	-0.83	+ 0.84
123	0.5669	0.5559	0.5451	-0.84	+ 0.85
124	0.5540	0.5432	0.5326	-0.85	+ 0.86
125	0.5415	0.5309	0.5205	-0.86	+ 0.87
126	0.5293	0.5189	0.5087	-0.87	+ 0.88
127	0.5175	0.5072	0.4972	-0.88	+ 0.89
128	0.5060	0.4959	0.4860	-0.89	+ 0.90
129	0.4948	0.4849	0.4752	-0.90	+ 0.90
130	0.4839	0.4741	0.4646	-0.91	+ 0.91
131	0.4732	0.4636	0.4542	-0.91	+ 0.92
132	0.4628	0.4534	0.4442	-0.92	+ 0.93
133	0.4527	0.4435	0.4344	-0.93	+ 0.94
134	0.4429	0.4338	0.4248	-0.94	+ 0.95
135	0.4333	0.4243	0.4155	-0.95	+ 0.96
136	0.4240	0.4152	0.4065	-0.96	+ 0.97
137	0.4149	0.4062	0.3977	-0.97	+ 0.98
138	0.4060	0.3975	0.3891	-0.98	+ 0.99
139	0.3974	0.3890	0.3808	-0.99	+ 1.00
140	0.3890	0.3808	0.3727	-1.00	+ 1.00
141	0.3808	0.3727	0.3647	-1.00	+ 1.01

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142	0.3728	0.3648	0.3570	-1.01 + 1.02
143	0.3650	0.3572	0.3495	-1.02 + 1.03
144	0.3574	0.3497	0.3421	-1.03 + 1.04
145	0.3500	0.3424	0.3350	-1.04 + 1.05
146	0.3428	0.3353	0.3280	-1.05 + 1.06
147	0.3358	0.3284	0.3212	-1.06 + 1.07
148	0.3289	0.3217	0.3146	-1.07 + 1.08
149	0.3223	0.3151	0.3081	-1.08 + 1.09
150	0.3157	0.3087	0.3018	-1.09 + 1.10